Boeing 747 Trajectory Tracking via Optimization, ODEs Solving, and State Estimation

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4071 Words Student ID:25144639R, 25145125R

Keywords: LQR, RK4, Bayesian estimation, MAP, Boeing 747, trajectory tracking

1st April, 2026

This project develops an integrated simulation framework for Boeing 747 circular-flight tracking in a 2D plane. The framework combines: (i) a linear quadratic regulator (LQR) for heading/bank inner-loop control, (ii) a fourth-order Runge-Kutta (RK4) integrator for nonlinear aircraft dynamics simulation, and (iii) a Gaussian Bayes/MAP estimator for state estimation under noisy measurements. Numerical experiments based on the project code show accurate path convergence and stable speed regulation, with final cross-track error close to zero and bounded bank angle throughout the simulation horizon. The report presents problem formulation, methodology, numerical results, and limitations. All LaTeX sources and Python code are openly available at https://github.com/edyou25/AAE6202.

1 Introduction

Circular holding patterns are among the most common maneuvers in commercial aviation, yet they demand tight coordination between heading control, bank-angle regulation, and speed maintenance. For a heavy transport aircraft like the Boeing 747, even small errors in any of these channels can accumulate over a long orbit, making the problem a useful testbed for control and estimation algorithms. This project takes the circular-tracking task as a concrete case study and builds an integrated simulation that couples three classical techniques: optimal feedback control, high-order numerical integration, and Bayesian state estimation.

The controller adopts a two-layer architecture. An outer guidance loop converts the geometric tracking error into a desired heading, while an inner loop applies Linear Quadratic Regulator (LQR) feedback to stabilize the heading and bank-angle errors[1,2]. LQR is chosen here not

Fig. 1. Representative Boeing 747 airframe image.

because it is the most advanced option, but because its closed-form gain computation and well-understood tuning behavior make it easy to diagnose when something goes wrong—an important consideration in a coursework setting. For discrete-time error dynamics

$$x_{k+1} = A_d x_k + B_d u_k,$$

the infinite-horizon LQR minimizes

$$J = \sum_{k=0}^{\infty} \left(x_k^\top Q x_k + u_k^\top R u_k \right),$$

with $Q \succeq 0$ and $R \succ 0$, leading to linear feedback $u_k = -K x_k$. In this project, LQR is used to stabilize heading and bank-angle errors around a circular-tracking reference.

Because the underlying aircraft dynamics are nonlinear, state propagation cannot rely on the linearized model alone. A fourth-order Runge–Kutta (RK4) integrator is used instead of a simpler Euler step. The additional function evaluations per time step are negligible on modern hardware, and the payoff—roughly two extra orders of accuracy for smooth problems—keeps the numerical error well below the control tolerance[3,4]. Section 4.6 provides a direct comparison that quantifies this advantage for the present B747 model.

To explore how sensor noise would affect a future closed-loop estimator, the simulation also includes a Gaussian Bayes filter that processes noisy position, heading, and speed measurements. Under linear-Gaussian assumptions the Bayes update reduces to a standard Kalman-style correction, whose posterior coincides with the Maximum A-Posteriori (MAP) estimate[5]. In the current implementation the filter runs in parallel with the controller but does not yet feed back into it; its main purpose is to record innovation statistics and covariance contraction so that a tighter integration can be attempted later.

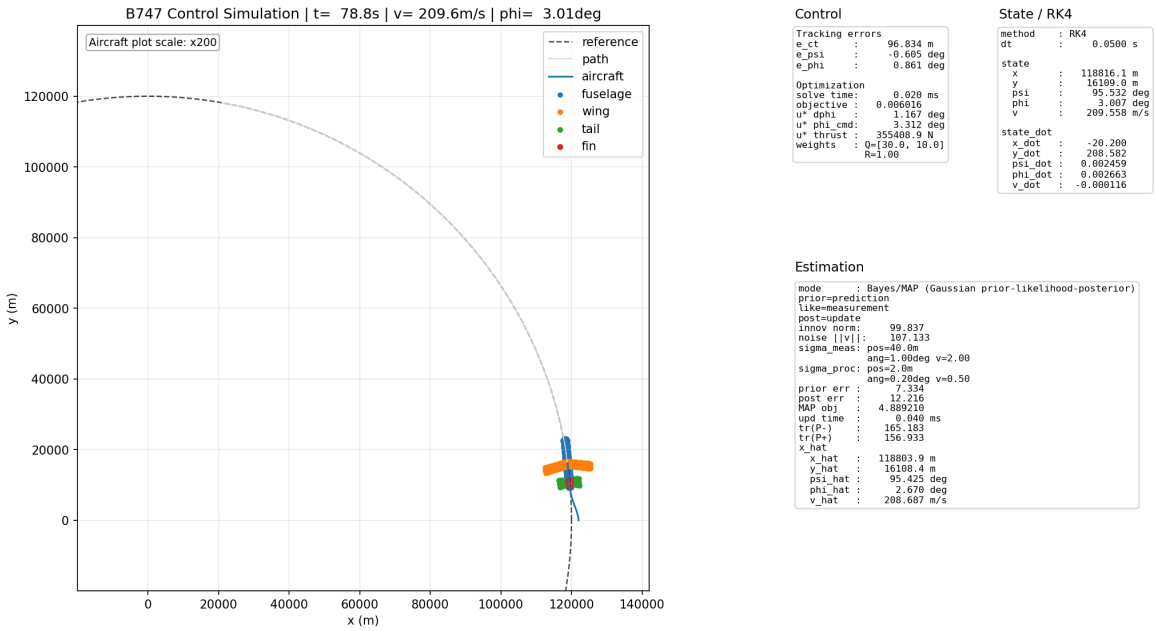

Fig. 2. Interactive overview of the B747 control simulation, including the trajectory view and diagnostic telemetry panels.

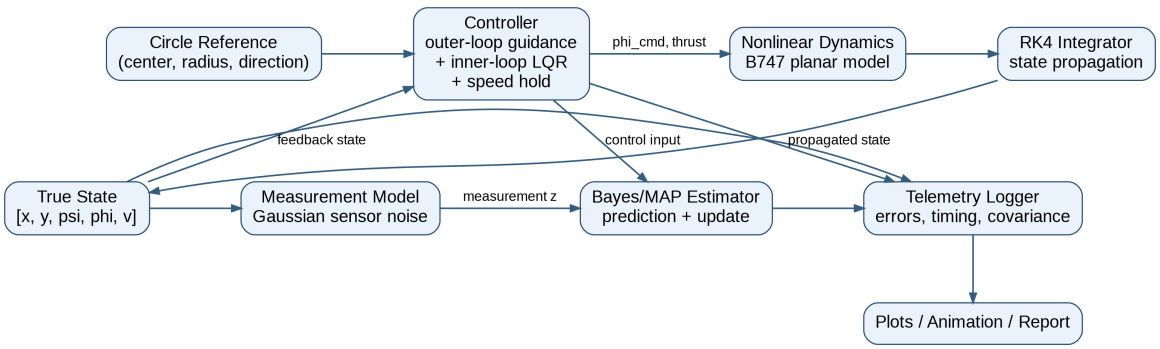

Fig. 3. System-level architecture of the integrated controller–simulation–estimator workflow.

The overall workflow aligns with the objectives of AAE6202, which emphasize optimization, numerical solution of differential equations, and uncertainty-aware modeling for aviation applications[6]. Figure 3 sketches the data flow among the guidance law, the LQR regulator, the RK4 propagator, and the Bayes filter, while Figure 2 shows a representative snapshot of the running simulation.

The remainder of this report is organized as follows. Section 2 states the tracking problem and the performance requirements. Section 3 details the nonlinear B747 model, the RK4 discretization, the layered controller, and the estimation algorithm. Section 4 presents numerical results including baseline performance, sensitivity studies, and a side-by-side RK4/Euler benchmark. Section 5 discusses limitations, and Section 6 concludes.

Figure 2 provides a compact visual overview of the integrated simulation environment used throughout this report.

2 Problem Statement

2.1 Objective

The task is to guide a Boeing 747 around a circular holding pattern while keeping speed nearly constant. Although real holding procedures involve altitude changes, wind corrections, and air-traffic constraints, this project isolates the horizontal tracking problem so that the interplay between guidance, control, and estimation can be studied without the complexity of a full six-degree-of-freedom model.

2.2 State, Input, and Reference

The aircraft state is described by five variables:

$$\mathbf{x} = [x,\ y,\ \psi,\ \phi,\ v]^\top,$$

where x and y are inertial coordinates, ψ is heading, ϕ is bank angle, and v is airspeed.

The control input is

$$\mathbf{u} = [\phi_{\text{cmd}},\ T]^\top,$$

where ϕ_{cmd} is the commanded bank angle and T is thrust.

The reference is a counter-clockwise circle of radius $R = 120\,\text{km}$ centred at the origin. This radius is deliberately large: at a cruise speed of $v_{\text{ref}} = 210\,\text{m/s}$ the required bank angle is only about 3.8°, well within the aircraft's comfortable envelope and far from any structural or passenger-comfort limit. Choosing a gentler turn allows the LQR to operate in a nearly linear regime and makes the simulation results easier to interpret. Tighter turns are examined briefly in the sensitivity study (Section 4.10).

2.3 Performance Requirements

The tracking system should satisfy the following requirements:

1. *Path convergence:* The signed cross-track error $e_{ct} = R - \|(x, y)\|$ should decay toward zero after an initial transient and remain small throughout steady-state flight.

2. *Heading and bank stability:* Neither the heading error nor the bank angle should exhibit sustained oscillation or overshoot beyond the design limits.

3. *Speed regulation:* Airspeed should stay close to $v_{\text{ref}} = 210\,\text{m/s}$, with deviations of no more than a few metres per second.

4. *Actuator feasibility:* Bank commands must respect the ±35° limit, and thrust must remain within the installed-engine capacity.

2.4 Constraints and Assumptions

The problem is simplified by the following assumptions:

1. Flight is modeled in a 2D plane.

2. Aerodynamic drag is represented by a quasi-steady model.

3. Control is updated discretely with $\Delta t = 0.05$ s, while the state dynamics are propagated continuously through numerical integration.

4. Measurement noise is Gaussian, and the estimator is used only for diagnostics in the current version.

3 Methodology

3.1 Parameter Reference

Table 1 lists every numerical value used in the simulation and states where it comes from. Aircraft mass, wing area, and engine thrust are taken from public Boeing and NASA documentation for the 747-400[7–9]. Aerodynamic coefficients follow standard drag-polar models with parameters quoted in university course notes[10,11]. Everything else—controller gains, estimator covariances, integration step size, and experiment settings—is a design choice made for this project and is not claimed to match any certified flight-control system.

3.2 Nonlinear B747 Planar Dynamics

The simulation treats the 747 as a point mass moving in a horizontal plane.

Position (x, y) evolves according to the kinematic relations

$$\dot{x} = v\cos\psi, \qquad \dot{y} = v\sin\psi,$$

$$\dot{\psi} = \frac{g\tan\phi}{v}, \qquad \dot{\phi} = \frac{\phi_{\text{cmd}} - \phi}{\tau_\phi}, \qquad \dot{v} = \frac{T - D}{m}.$$

This simple model captures the fact that the roll autopilot cannot change bank instantaneously; two seconds is a reasonable order of magnitude for a large transport aircraft, though not a certified value.

The aerodynamic drag model is

$$D = \frac{1}{2}\rho v^2 S C_D, \qquad C_D = C_{D0} + kC_L^2, \qquad C_L = \frac{2mg}{\rho v^2 S\cos\phi}.$$

Aircraft parameters are summarised in Table 1.

Table 1
The main numerical parameters used in the report.

Parameter group	Values used in this report	Source of the values
Aircraft physical model	$m = 396{,}890\,\mathrm{kg}$, $S = 510.97\,\mathrm{m}^2$, $C_{D0} = 0.022$, $k = 0.048$	B747-400 maximum takeoff weight and installed-engine family data from Boeing; wing reference area from NASA's 747-400 PCA model; C_{D0} from MIT notes quoting Boeing drag data; k inferred from $k = 1/(\pi e AR)$ using sourced geometry and a Raymer-type span-efficiency estimate[7,8,10,11]
Actuation and envelope assumptions	$\tau_\phi = 2.0\,\mathrm{s}$, $\lvert\phi\rvert \le 35°$, $v \in [120, 320]\,\mathrm{m/s}$	Simplified simulation assumptions introduced for numerical robustness and controller design; these are not claimed as manufacturer-issued 747 limits
Installed thrust model	$T \in [0, 1.126 \times 10^6]\,\mathrm{N}$	Based on four PW4062-class engines rated at about 63,300 lbf each on 747-400 variants[7,9]
Nominal mission setup	$R = 120{,}000\,\mathrm{m}$, $v_{\mathrm{ref}} = 210\,\mathrm{m/s}$, $\Delta t = 0.05\,\mathrm{s}$, $t_{end} = 800\,\mathrm{s}$	Coursework simulation choices for the baseline circular-tracking experiment
Initial condition	$[R + 2000, 0, 96°, 0°, 210]$	Deliberate off-circle initial condition chosen to create a visible convergence transient
Controller tuning	$Q = \mathrm{diag}(30, 10)$, $R = [1]$, $k_v = 120000$, $k_{ct} = 1/3500$, heading-correction clip $= \pm 25°$	Project controller-design choices for the nominal case study
Estimator covariances	Process std $(2, 2, 0.2°, 0.2°, 0.5)$, measurement std $(40, 40, 1°, 1°, 2)$, initial std $(50, 50, 2°, 2°, 3)$	Project estimator settings for diagnostic uncertainty visualization
Sensitivity sweeps	$\Delta t \in \{0.02, 0.05, 0.10, 0.20\}\,\mathrm{s}$, $R \in \{80, 120, 160, 200\}\,\mathrm{km}$, noise scale $\in \{0.5, 1, 2, 4\}$, three LQR weight sets	Experiment-design choices for robustness and sensitivity analysis
RK4/Euler benchmark	$\Delta t_{\mathrm{ref}} = 0.05\,\mathrm{s}$, $\Delta t_{\mathrm{test}} = 0.5\,\mathrm{s}$, $T_{\mathrm{sim}} = 100\,\mathrm{s}$, $\phi_{\mathrm{cmd}} = 15° + 10° \sin(2\pi t/20)$	Independent numerical benchmark settings chosen to compare integrators under oscillatory bank input

The mass $m = 396{,}890\,\mathrm{kg}$ corresponds to the 747-400 maximum take-off weight, and the wing area $S = 510.97\,\mathrm{m}^2$ is taken from NASA's Propulsion Controlled Aircraft benchmark[8]. Thrust is limited to $1.126 \times 10^6\,\mathrm{N}$, consistent with four PW4062-class engines[9]. Bank angle is clipped to $\pm 35°$ and airspeed to the interval $[120, 320]\,\mathrm{m/s}$ to keep the simulation within a physically reasonable envelope.

3.3 RK4 Discretization

For each simulation step, with constant input over $[t_k, t_{k+1}]$, RK4 evaluates

$$k_1 = f(x_k, u_k), \qquad k_2 = f\left(x_k + \frac{\Delta t}{2} k_1, u_k\right),$$

$$k_3 = f\left(x_k + \frac{\Delta t}{2} k_2, u_k\right), \qquad k_4 = f\left(x_k + \Delta t\, k_3, u_k\right),$$

and updates the state by

$$x_{k+1} = x_k + \frac{\Delta t}{6}(k_1 + 2k_2 + 2k_3 + k_4).$$

For smooth right-hand sides RK4 has local truncation error $O(\Delta t^5)$, which translates to a global error of $O(\Delta t^4)$ over a fixed time horizon. At the baseline step size $\Delta t = 0.05$ s, Section 4.6 shows that switching to forward Euler at the same step size would inflate the trajectory error by roughly an order of magnitude.

3.4 Circular Tracking Controller

The controller is organised in two nested loops, following standard practice for aircraft guidance[2]. The outer loop converts cross-track error into a desired heading; the inner loop regulates heading and bank with LQR feedback.

First, an outer-loop geometry-based guidance law computes current radius, circle tangent heading, and signed cross-track error. A bounded heading correction is added:

$$\psi_{\text{des}} = \psi_{\text{tangent}} + \text{clip}\left(\arctan\left(k_{ct}(-e_{ct})\right), \pm 25^\circ\right).$$

Second, a feedforward bank command for circular motion is generated from

$$\phi_{ff} = \arctan\left(\frac{v_{\text{ref}}^2}{gR}\right),$$

with the correct sign for the direction of turn and with saturation at the bank-angle limit.

Third, an inner-loop LQR regulates heading and bank-angle errors,

$$e_\psi = \psi - \psi_{\text{des}}, \qquad e_\phi = \phi - \phi_{ff}.$$

Using the linearized error dynamics

$$\dot{e}_\psi = \frac{g}{v_{\text{ref}}} e_\phi, \qquad \dot{e}_\phi = -\frac{1}{\tau_\phi} e_\phi + \frac{1}{\tau_\phi} u,$$

where $u = \Delta\phi_{\text{cmd}}$, the code forms a discrete approximation

$$A_d \approx I + \Delta t A, \qquad B_d \approx \Delta t B,$$

solves the discrete Riccati equation, and applies

$$\Delta\phi_{\text{cmd}} = -K[e_\psi,\, e_\phi]^\top, \qquad \phi_{\text{cmd}} = \phi_{ff} + \Delta\phi_{\text{cmd}}.$$

The speed loop uses

$$T = D_{\text{guess}} + k_v(v_{\text{ref}} - v),$$

followed by clipping to thrust bounds. The feedforward-turn formula comes from coordinated-turn kinematics, while the numerical values $k_v = 120000$, $k_{ct} = 1/3500$, and the $\pm 25°$ heading-correction clip are baseline controller-tuning choices for this case study[2].

Default LQR weighting in `ControlConfig` is

$$Q = \text{diag}(30,\, 10), \qquad R = [1].$$

These weights are not taken from a handbook table; they are the baseline tuning adopted for the nominal simulation and the later sensitivity study.

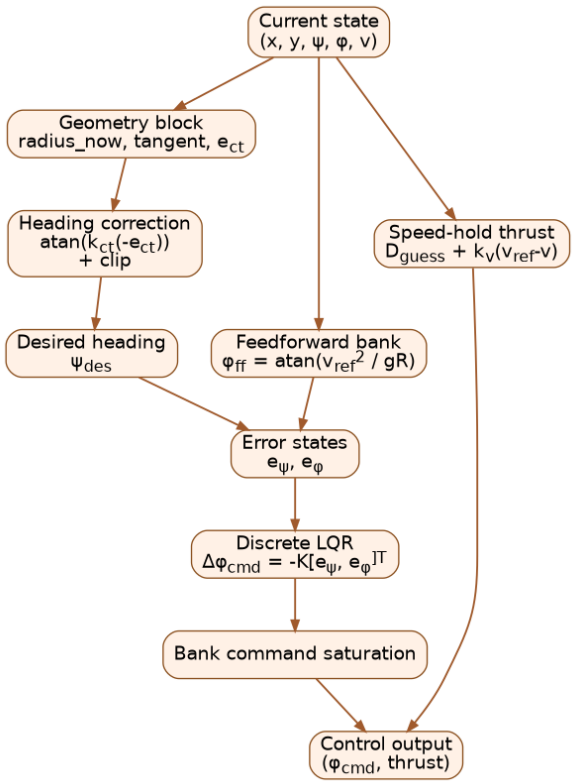

Fig. 4. Detailed control-logic flow from geometric error to saturated command output.

Figure 4 makes the layered design explicit. Geometric guidance computes the desired heading, coordinated-turn physics provides a feedforward bank angle, and LQR then stabilizes the residual angular errors. This decomposition is easier to interpret and tune than a monolithic controller, which is one reason it works well in an educational setting.

3.5 Gaussian Bayes/MAP Estimator

Although the controller currently uses the true state, a Gaussian Bayes filter is run in parallel to study how sensor noise would propagate through the estimation loop. The filter follows the standard predict–update cycle.

The state estimate is propagated through the same RK4 integrator used for the true dynamics:

$$x_k^- = f_{RK4}(x_{k-1}^+, u_{k-1}), \qquad P_k^- = P_{k-1}^+ + Q.$$

Update:

$$K_k = P_k^-(P_k^- + R)^{-1},$$

$$x_k^+ = x_k^- + K_k(z_k - x_k^-), \qquad P_k^+ = (I - K_k)P_k^-.$$

The MAP objective evaluated at the posterior state is

$$\mathcal{L}_{MAP} = \frac{1}{2}(x_k^+ - x_k^-)^\top (P_k^-)^{-1}(x_k^+ - x_k^-) + \frac{1}{2}(z_k - x_k^+)^\top R^{-1}(z_k - x_k^+).$$

The noise settings are:

1. Process standard deviation: $(2,\ 2,\ 0.2^\circ,\ 0.2^\circ,\ 0.5)$
2. Measurement standard deviation: $(40,\ 40,\ 1^\circ,\ 1^\circ,\ 2)$
3. Initial standard deviation: $(50,\ 50,\ 2^\circ,\ 2^\circ,\ 3)$

These covariance surrogates are project-level estimator settings for diagnostic visualization rather than quantities identified from flight-test data.

3.6 Simulation Workflow

At each simulation step, the following sequence is executed:

1. Compute control action and control diagnostics.
2. Generate a noisy measurement.
3. Run the estimator update.
4. Propagate the true state with RK4.
5. Log telemetry for analysis and visualization.

3.7 Software Architecture and Data Flow

From a software-engineering perspective, the project is organized as a compact closed-loop pipeline. The geometric reference is supplied to the controller, the controller generates bank-angle and thrust commands, the nonlinear B747 dynamics are propagated by RK4, and noisy measurements are processed by the Gaussian Bayes/MAP estimator. At the same time, telemetry signals such as tracking errors, timing, innovation norms, covariance traces, and state derivatives are collected for post-processing and report generation. This structure is intentionally modular: each mathematical part of the coursework corresponds directly to one code module.

For reproducibility, the implementation structure is also summarized through a curated project tree. The root directory contains the main simulation modules, including controller design, nonlinear dynamics, estimation, reporting, visualization, and the executable entry point. The `tests/` folder stores unit tests for each functional module. The `docs/` folder contains the written technical notes and report drafts. The `data/` folder stores runtime animations, summary metrics, and test outputs. The `latex/` folder contains the final report source, bibliography, class file, compiled PDF artifacts, and a dedicated `assets/` directory that centralizes every Python-generated figure and Graphviz source referenced by the manuscript.

Curated Project Tree

```
AAE6202
|-- controller.py
|-- dynamics.py
|-- estimation.py
|-- reporting.py
|-- extended_reporting.py
|-- run.py
|-- visual.py
|-- env.yaml
|-- README.md
|-- README_en.md
|-- assets
|   |-- 747.png
|   \-- ...
|-- data
|   |-- circle_flight_animation.gif
|   |-- report_ext
|   |   \-- report_metrics.json
|   \-- test_outputs
|-- docs
|   |-- estimation.md
|   |-- lqr.md
|   |-- project_report.md
|   |-- rk.md
|   |-- unit_test_report.md
|-- latex
|   |-- abstract.tex
|   |-- Academic.cls
|   |-- references.bib
```

```
|   |-- wordcount.py
|   |-- _main.tex
|   |-- assets
|   |   |-- controller_logic.dot
|   |   |-- controller_logic.png
|   |   |-- dt_sensitivity.png
|   |   \-- ...
|   |-- output
|   |   |-- _main-words.tex
|   |   \-- _main.pdf
|   \-- template
|       |-- abstract.tex
|       |-- Academic.cls
|       |-- references.bib
|       \-- _main.tex
|-- tests
|   |-- test_controller.py
|   |-- test_dynamics.py
|   |-- test_estimation.py
|   |-- test_eular_rk.py
|   |-- test_reporting.py
|   |-- test_run.py
|   \-- test_visual.py
\-- .git
```

Figure 3 emphasizes that the controller and estimator use the same dynamic context but serve different roles. The controller shapes trajectory-tracking behavior, whereas the estimator is currently diagnostic and does not yet close the feedback loop. This separation is useful in a course project because it allows control performance and estimation behavior to be studied independently before tighter integration is attempted.

Environment File (`env.yaml`)

```yaml
name: aae6202
channels:
  - conda-forge
  - defaults
dependencies:
  - python=3.11
  - numpy
  - matplotlib
  - pillow
  - pip
```

4 Results

4.1 Experiment Setup

The baseline experiment runs for $t_{\text{end}} = 800\,\text{s}$ with a control/logging interval of $\Delta t = 0.05\,\text{s}$, giving 16 000 time steps—long enough for the aircraft to complete roughly one-third of the 120 km orbit. The initial condition places the aircraft 2 km outside the reference circle, heading nearly tangent to it at $\psi_0 = 96°$, with zero initial bank and the target speed of 210 m/s. This deliberate offset creates a visible convergence transient before steady-state tracking is reached.

4.2 Tracking and Regulation Performance

Table 2 reports the nominal closed-loop metrics.

Table 2
Tracking and regulation performance metrics

Metric	Value
Final signed cross-track error e_{ct}	0.218 m
Final $\lvert e_{ct}\rvert$	0.218 m
Overall RMSE of e_{ct} (0–800 s)	269.24 m
MAE of e_{ct}	60.64 m
RMSE of e_{ct} (100–800 s)	5.56 m
Settling time to $\lvert e_{ct}\rvert < 500$ m	27.8 s
Settling time to $\lvert e_{ct}\rvert < 100$ m	78.35 s
Heading error RMSE	2.31°
Max $\lvert\phi\rvert$	27.67°
Max $\lvert\phi_{\text{cmd}}\rvert$	35.0°
Command saturation ratio	0.356%
Mean speed	209.724 m/s
Speed RMSE to $v_{\text{ref}} = 210$	0.276 m/s

The overall cross-track RMSE is dominated by the intentional initial offset, but after the transient the residual radial error is small. Bank angle remains within limits, command saturation is rare, and speed tracking stays tight around 210 m/s.

Figure 5 confirms smooth decay of the tracking errors, close agreement between commanded and realized bank angle, and well-behaved thrust demand.

4.3 Computational Efficiency

Table 3 shows controller and estimator timing.

Both runtimes are negligible relative to the 50 ms control interval, so the framework is comfortably fast enough for the present simulation setting.

Figure 6 shows that the timing distributions are compact and do not exhibit problematic long tails.

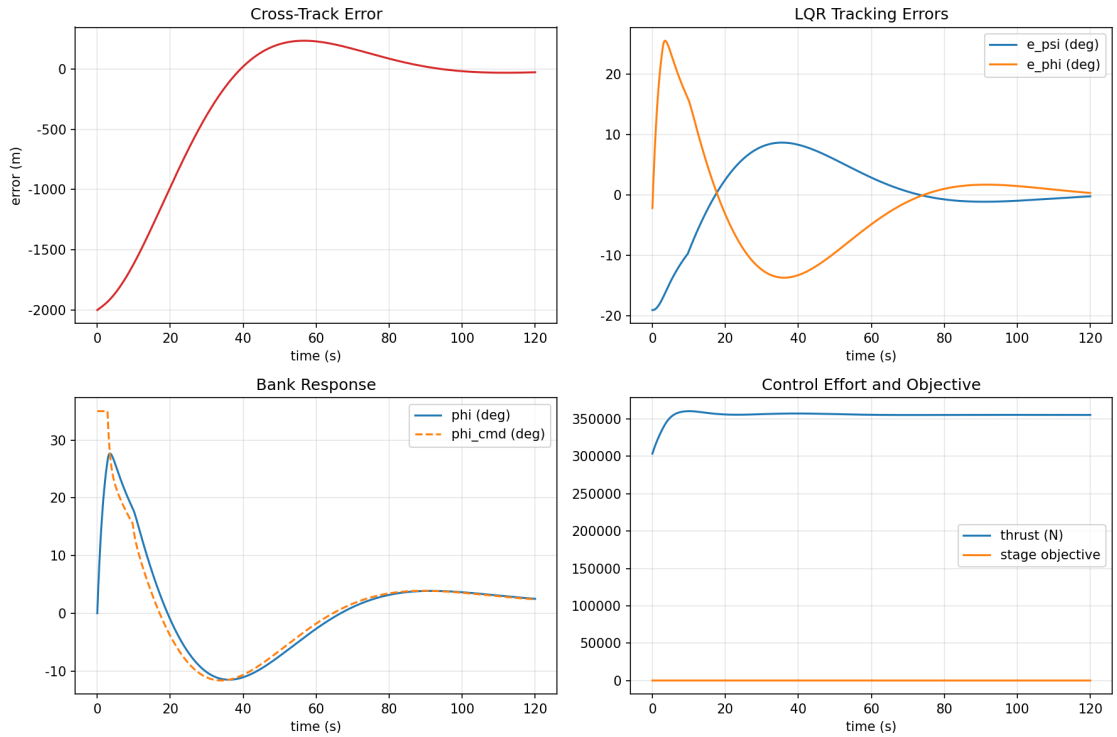

Fig. 5. Detailed control figure: tracking errors, bank response, and control effort in the nominal case.

Table 3
Computation time metrics

Metric	Value
Mean controller computation time	0.0165 ms
95th percentile controller time	0.0176 ms
Mean estimator update time	0.0344 ms
95th percentile estimator time	0.0365 ms

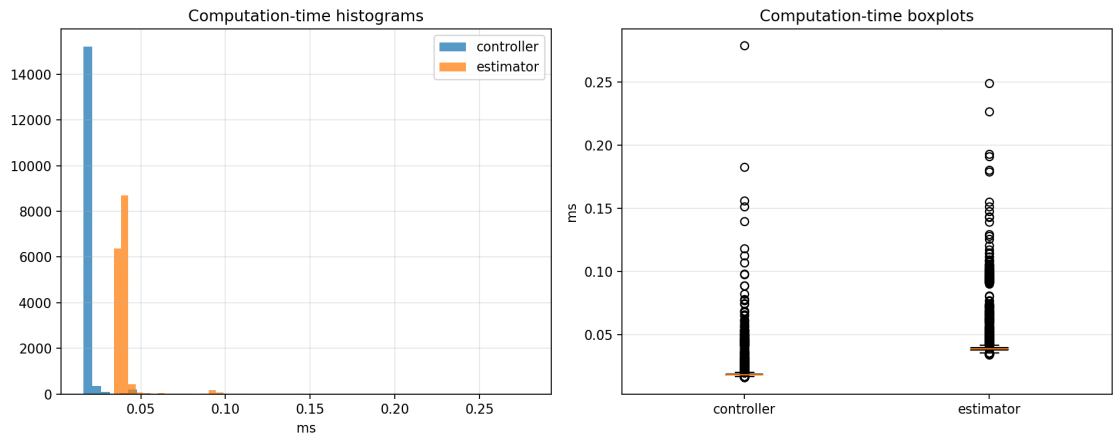

Fig. 6. Distribution of controller and estimator computation times.

4.4 Estimation Diagnostics

Table 4 summarizes the nominal estimation statistics.

Table 4
Estimator diagnostic metrics

Metric	Value
Mean prior error norm $\|x^- - x\|$	8.380
Mean posterior error norm $\|x^+ - x\|$	8.379
Fraction of steps where posterior improves error	50.62%
Mean innovation norm $\|z - x^-\|$	50.67
Mean trace reduction $\mathrm{tr}(P^-) \to \mathrm{tr}(P^+)$	5.15%

The covariance contracts after each update, but the posterior mean improves only slightly over the prior because process and measurement uncertainties are of similar scale. Since the controller still uses the true state, these results mainly quantify estimator consistency rather than closed-loop benefit.

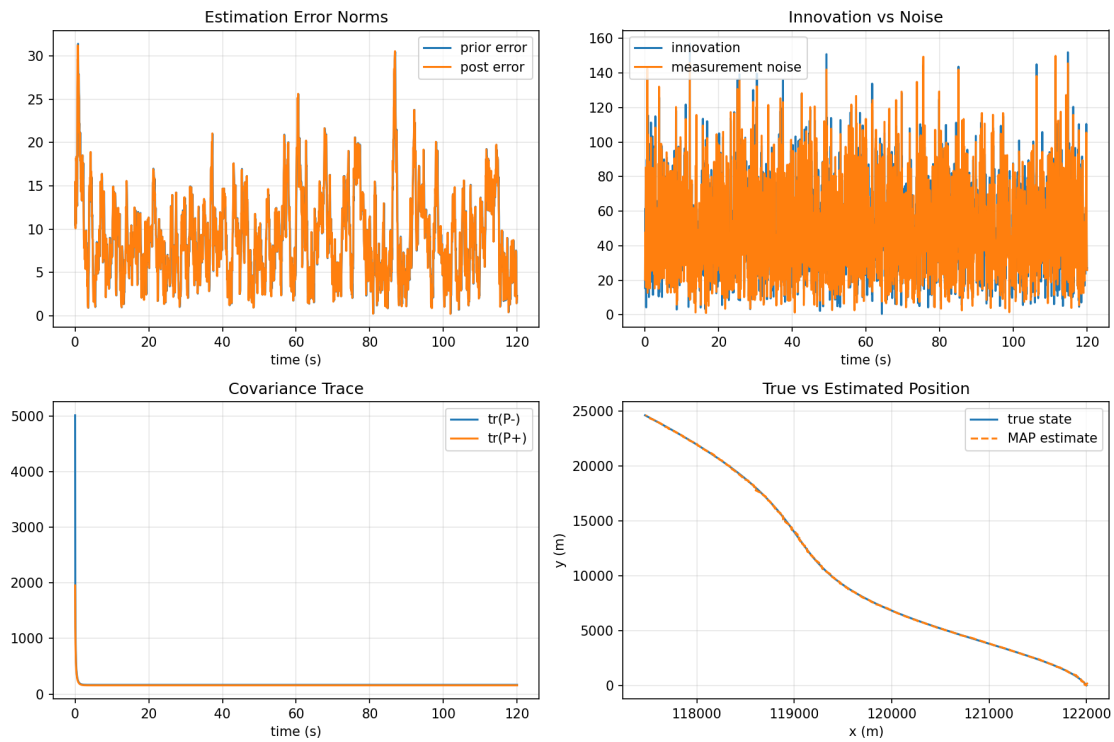

Fig. 7. Compact estimation summary: error norms, innovation, covariance trace, and true-versus-estimated position.

Figure 7 visualizes prior/posterior errors, innovation magnitude, covariance reduction, and estimated versus true position in the nominal case.

Figure 8 adds the covariance-trace and MAP-objective detail behind the scalar estimator metrics.

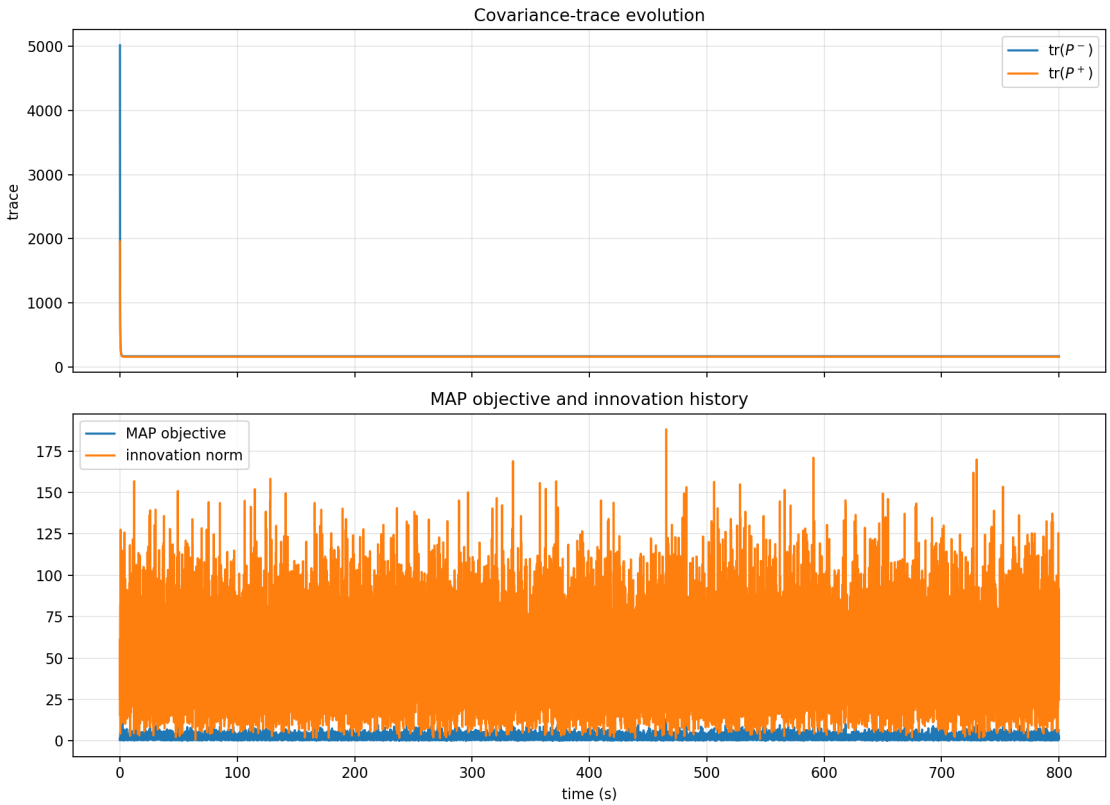

Fig. 8. Detailed estimation diagnostics: covariance traces, MAP objective, and innovation history.

4.5 Trajectory and State Histories

The baseline trajectory starts outside the target circle and converges smoothly to stable circular motion. The heading, bank-angle, and speed states remain bounded and physically reasonable throughout the run.

Figure 9 provides a compact global view of the nominal closed-loop behavior before the detailed state plots are examined.

Figure 10 shows that the nominal trajectory follows the reference circle cleanly and that the translational, angular, and speed channels remain smooth throughout the maneuver.

4.6 RK4/Euler Cross-Check

To justify the choice of integrator, a separate benchmark propagates the same B747 model with both RK4 and forward Euler at a deliberately coarse step size of $\Delta t = 0.5$ s—ten times larger than the baseline. The commanded bank follows a sinusoidal profile so that the nonlinear heading dynamics are exercised continuously.

After 100 s of simulated flight, Euler accumulates a position error of roughly 44 m while RK4 stays within 3 m of the fine-step reference—an order-of-magnitude improvement for four times the function evaluations per step. Figure 19 breaks the error down by state component; the

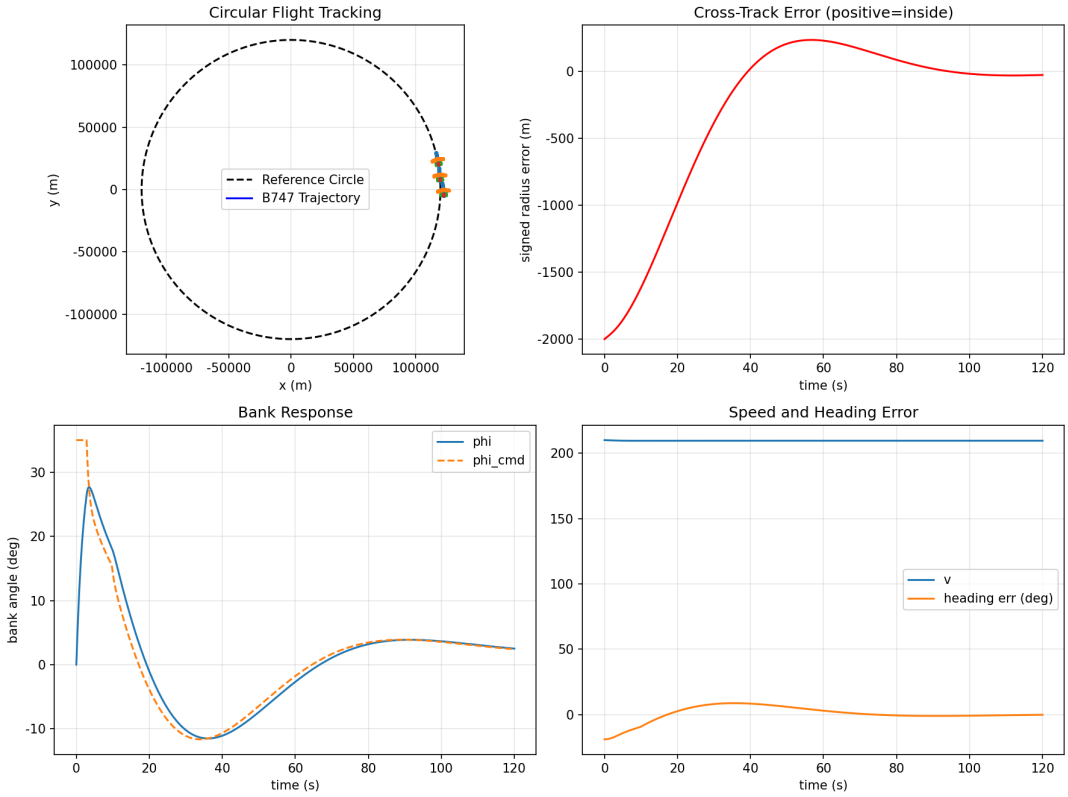

Fig. 9. Overview of circular tracking performance generated from the project code.

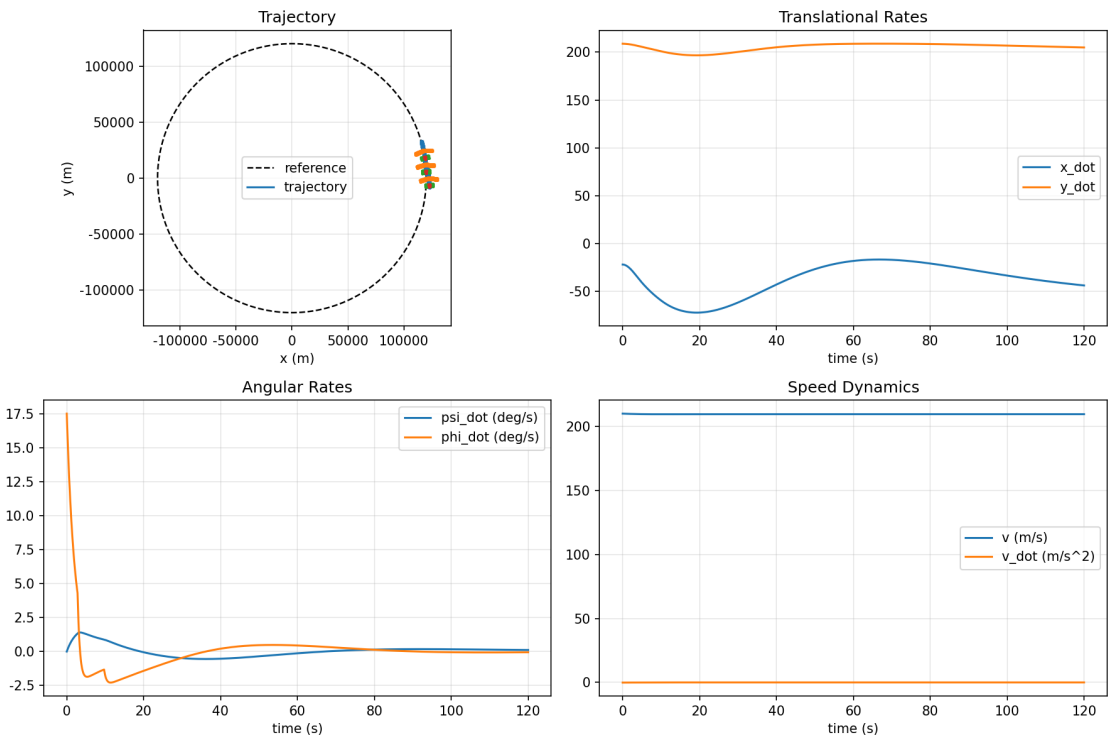

Fig. 10. Detailed dynamics figure: trajectory, translational rates, angular rates, and speed behavior.

advantage is most pronounced in heading and position, where the $\tan\phi$ nonlinearity in $\dot{\psi}$ causes Euler's local errors to compound.

At the baseline step of 0.05 s both methods would give acceptable accuracy, but RK4's larger stability region also allows occasional coarser steps without divergence—a useful margin if the simulation is ever extended to stiffer dynamics or variable-step integration.

4.7 Transient Behavior in the First 120 Seconds

Figure 11 isolates the first 120 s, where the aircraft removes the initial radial and angular offsets.

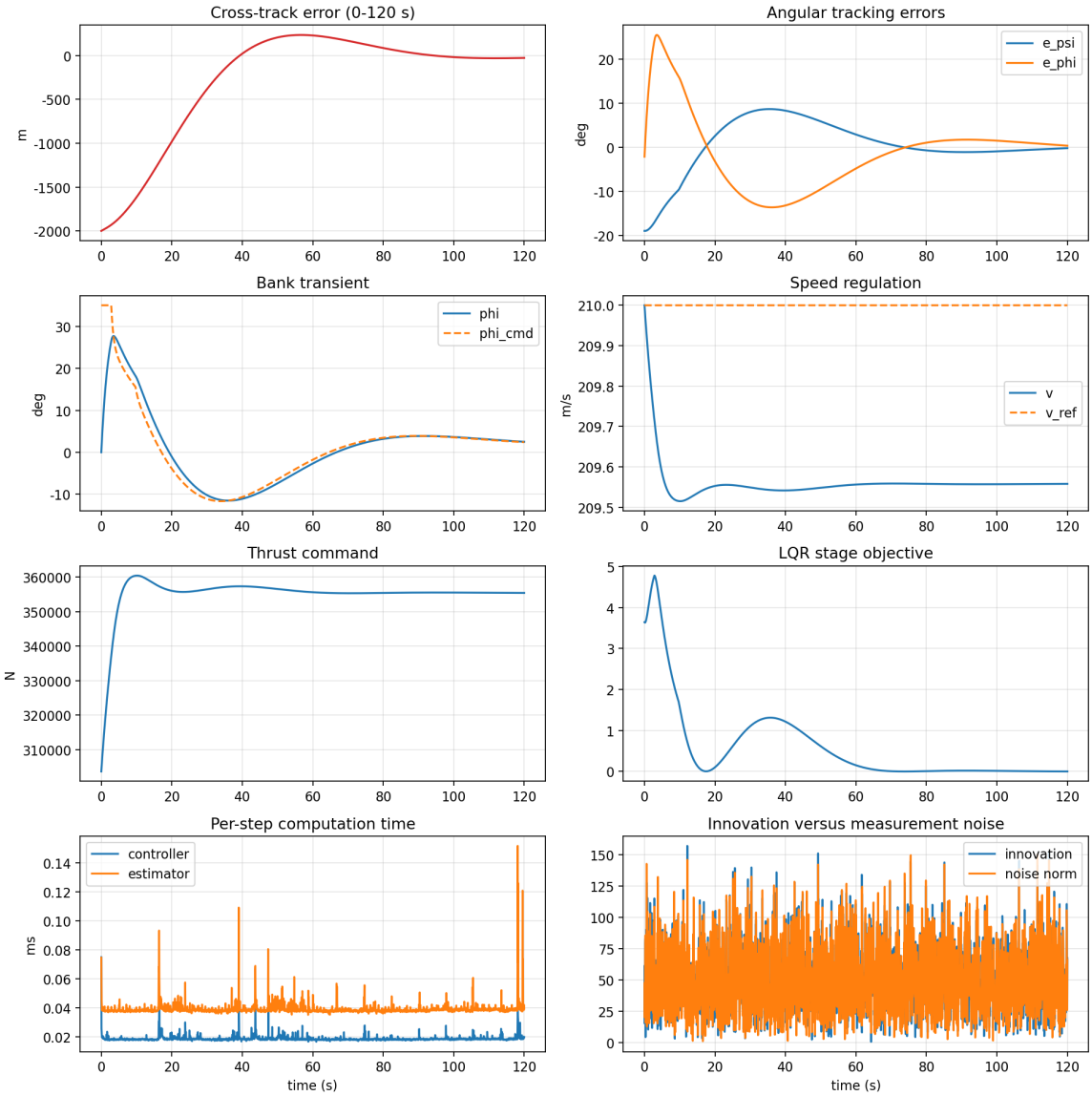

Fig. 11. Detailed transient response during the first 120 seconds of the baseline simulation.

The transient is smooth: cross-track error decreases rapidly, bank angle follows the command with first-order lag, speed remains close to 210 m/s, and computation times stay negligible relative to the sampling interval.

4.8 State-Estimation Consistency and Error Geometry

Figure 12 compares each true state with its posterior estimate, while Figure 13 shows the controller error geometry and bank-angle distributions.

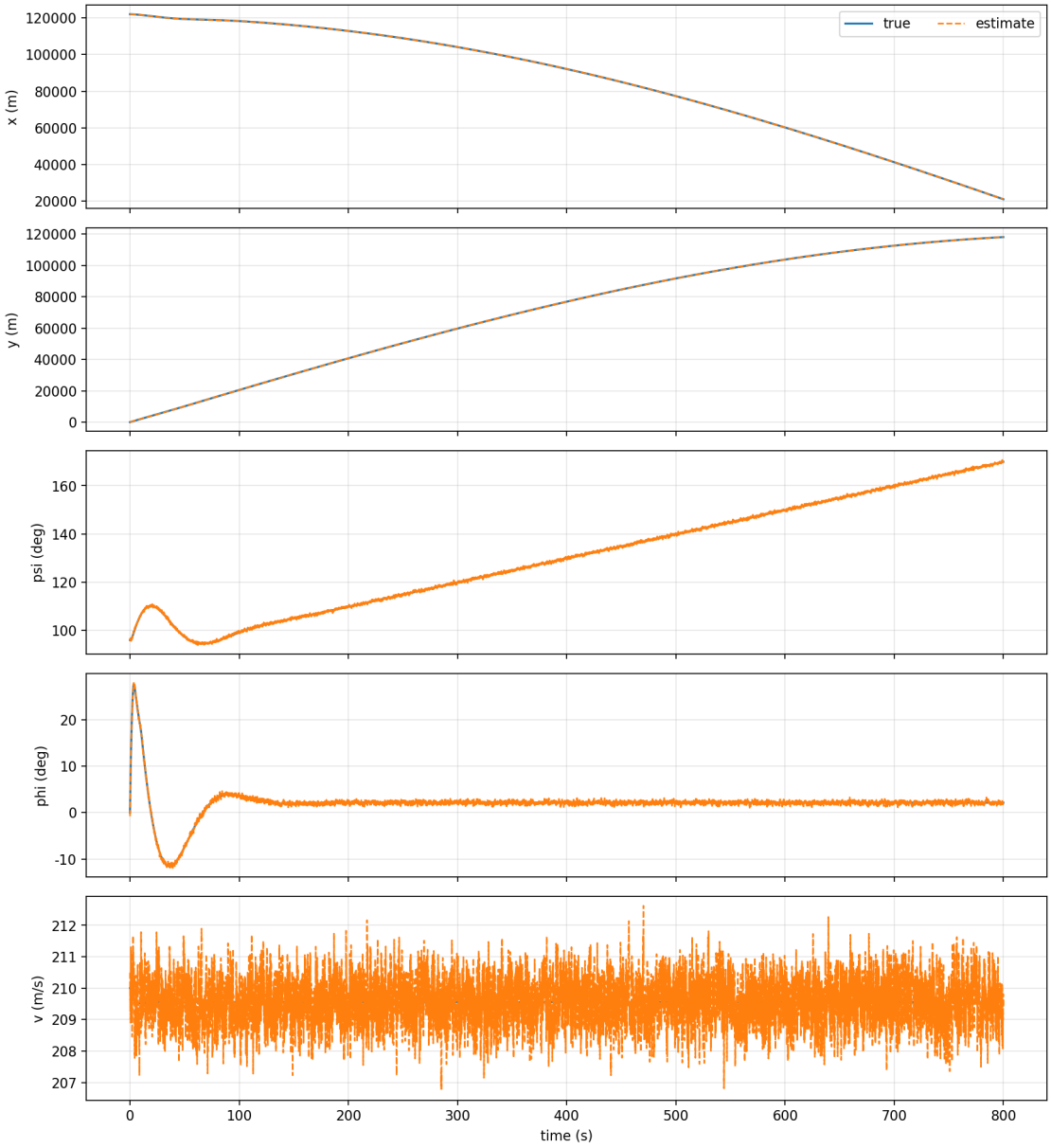

Fig. 12. True state and posterior estimate for all five state components.

The true and estimated states remain close after the initial transient, consistent with the scalar estimation metrics reported earlier. The phase portrait also shows that the controller drives the joint error state toward the origin without large oscillatory loops.

4.9 Sensitivity to Integration Time Step

Figure 14 compares closed-loop metrics for $\Delta t = 0.02, 0.05, 0.10$, and $0.20\,\mathrm{s}$ under otherwise identical settings.

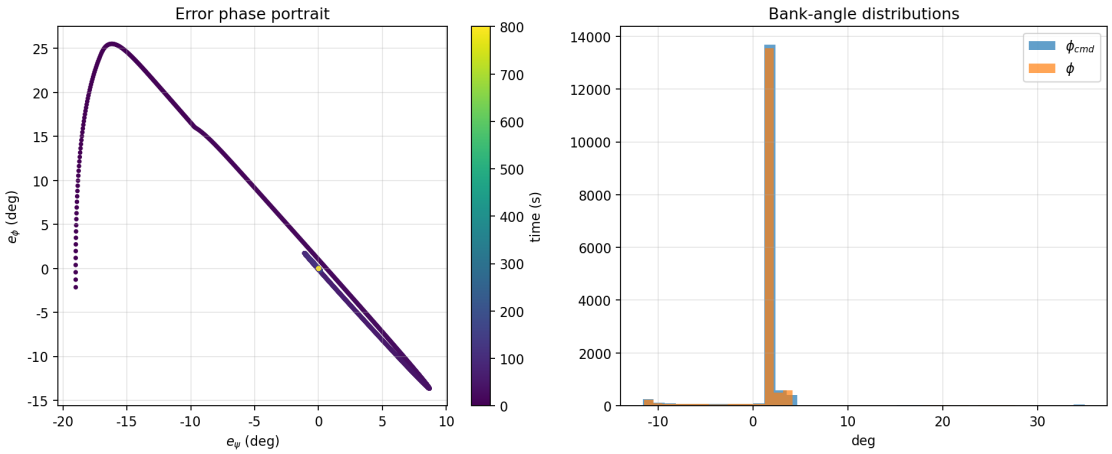

Fig. 13. Error phase portrait and bank-angle distributions in the baseline case.

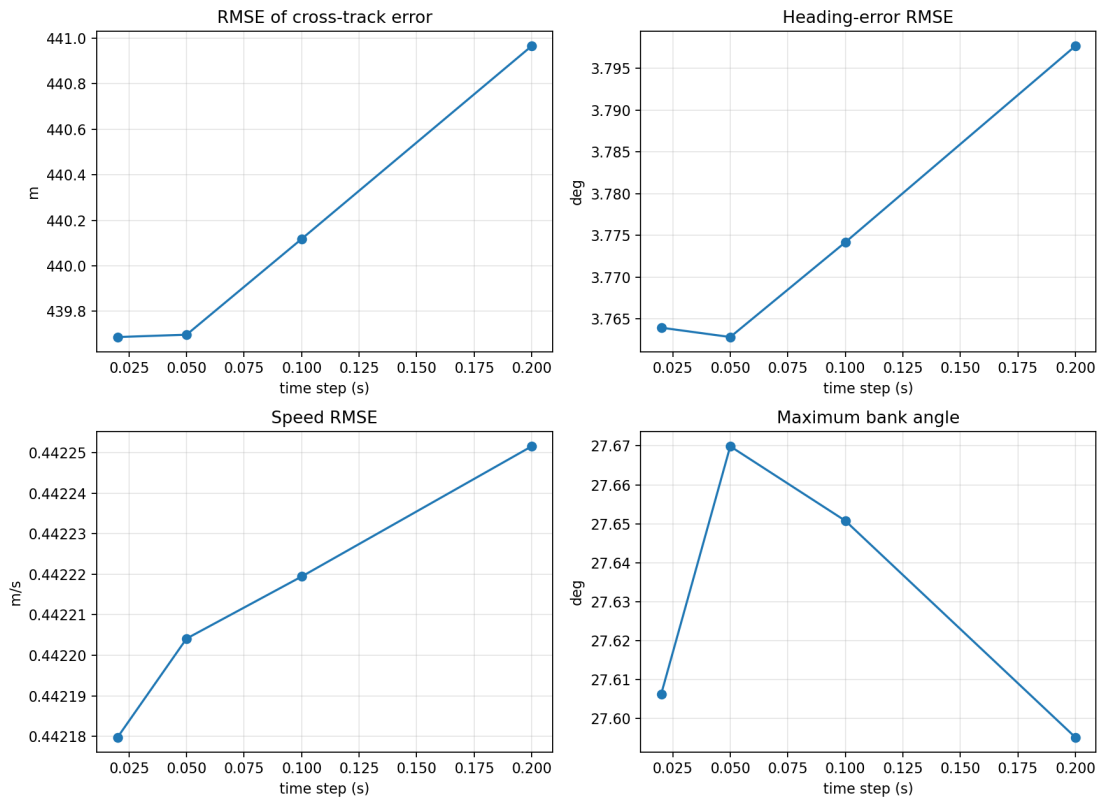

Fig. 14. Sensitivity of tracking and regulation metrics to integration time step.

Table 5
Selected metrics for the time-step sensitivity study

Δt (s)	RMSE(e_{ct}) (m)	Heading RMSE (deg)	Speed RMSE (m/s)	Max $\lvert\phi\rvert$ (deg)
0.02	439.63	3.77	0.2762	27.61
0.05	439.64	3.76	0.2762	27.67
0.10	440.06	3.78	0.2762	27.65
0.20	440.91	3.80	0.2762	27.60

Table 5 and Figure 14 show only mild variation across the tested step sizes. The baseline choice $\Delta t = 0.05\,\mathrm{s}$ is therefore sufficiently conservative for stable and accurate closed-loop simulation.

4.10 Sensitivity to Reference-Circle Radius

Figure 15 compares four reference radii between 80,000 m and 200,000 m.

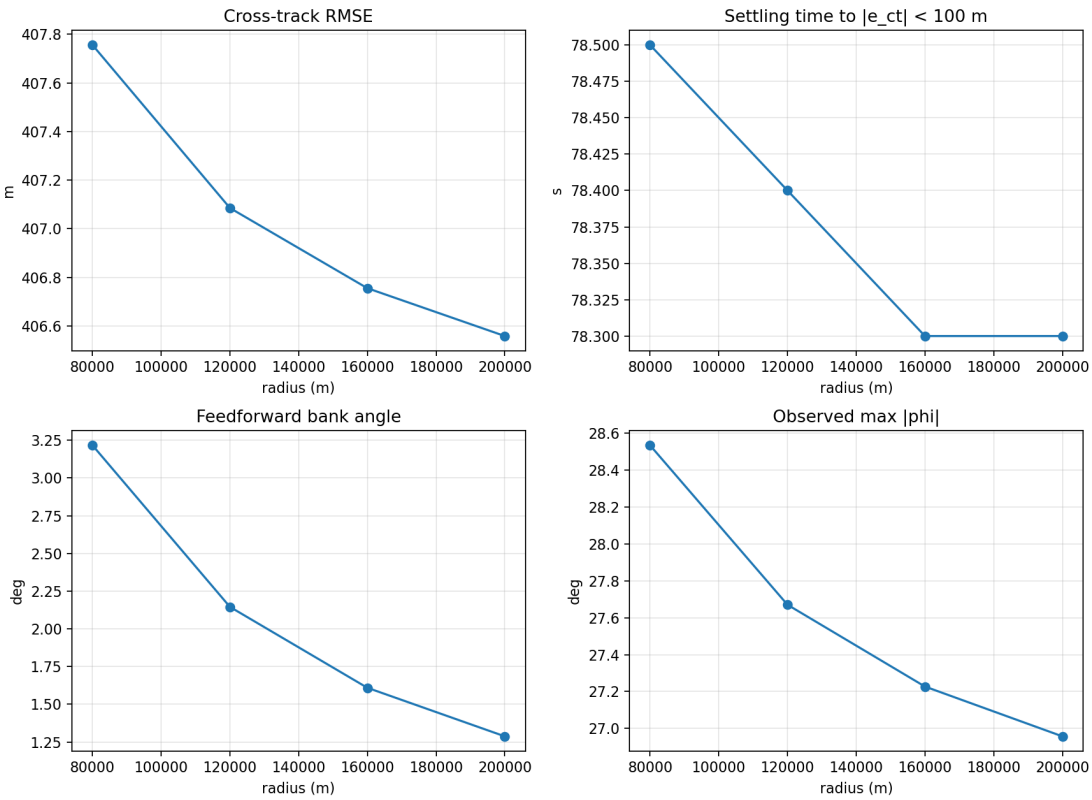

Fig. 15. Sensitivity of performance and commanded curvature to reference-circle radius.

As the radius increases, the feedforward and observed bank angles decrease monotonically, while tracking RMSE changes only modestly. The controller therefore generalizes well across a useful range of path curvatures.

Figure 16 shows that all tested cases converge toward their respective circles without pathological behavior.

4.11 Sensitivity to Measurement Noise

Figure 17 scales the measurement-noise covariance from 0.5 to 4.0 times the baseline value.

Tracking metrics remain nearly unchanged because control does not yet use the estimate, whereas estimation errors and innovation norms increase clearly with noise level. The sweep therefore isolates estimator sensitivity without confounding it with controller behavior.

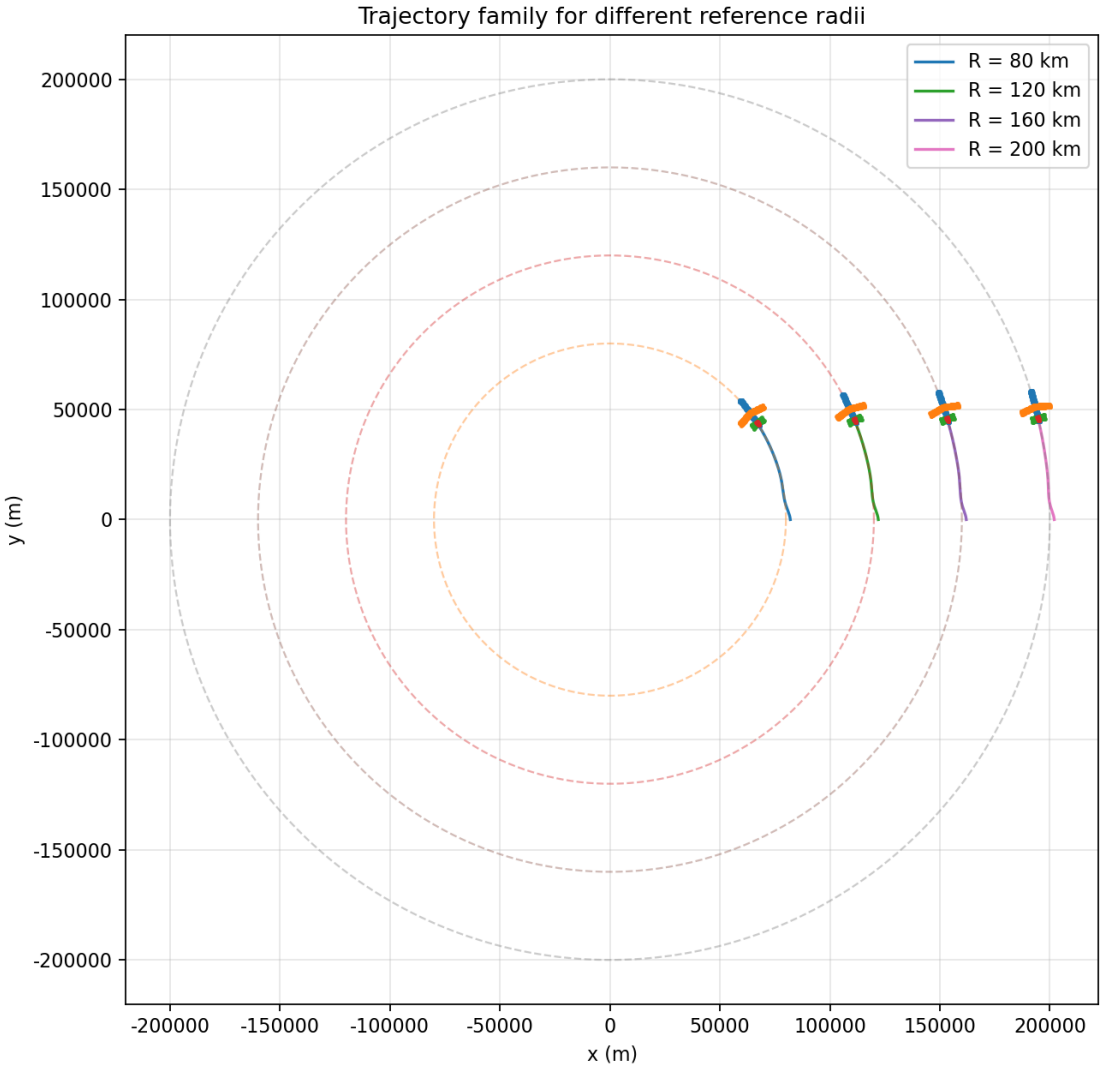

Fig. 16. Family of closed-loop trajectories under different reference-circle radii.

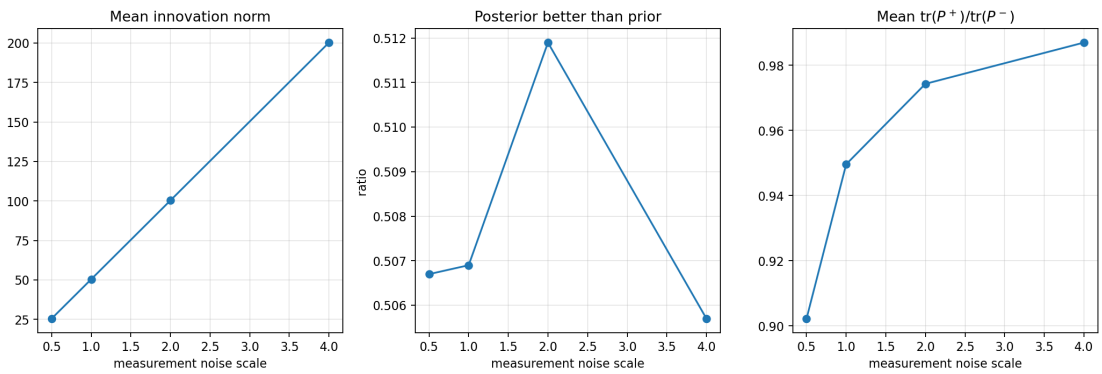

Fig. 17. Estimator sensitivity to measurement-noise scaling.

4.12 Sensitivity to LQR Weight Selection

Figure 18 compares conservative, baseline, and aggressive LQR weights.

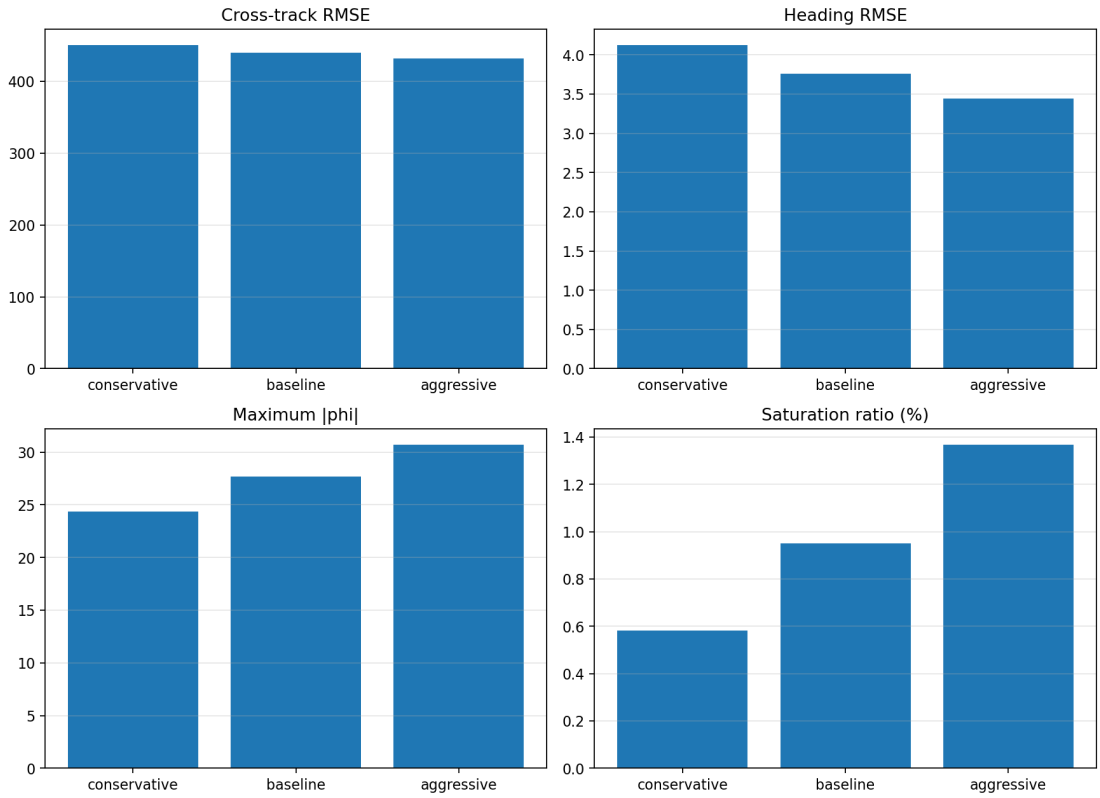

Fig. 18. Effect of LQR weight selection on tracking quality and actuator usage.

The aggressive design improves tracking but increases peak bank angle and saturation, whereas the conservative design reduces control effort at the expense of weaker correction. The baseline choice remains the best compromise between responsiveness and actuator margin.

4.13 Recreated RK4-versus-Euler Benchmark

Figure 19 recreates the supplementary RK4/Euler benchmark using the project dynamics implementation.

The recreated benchmark matches the independent script closely: RK4 is about 16.46× better than Euler in final total state error and about 12.68× better in mean total state error. This consistency reinforces the numerical advantage of RK4 for the present nonlinear dynamics model.

5 Discussion

The combined results show that the controller-estimator architecture is stable, computationally light, and accurate enough for circular tracking in the present nonlinear planar model. Outer-loop geometric correction removes large radial offsets effectively, inner-loop LQR regulates heading and bank dynamics well, and RK4 keeps propagation error from dominating the closed-loop response.

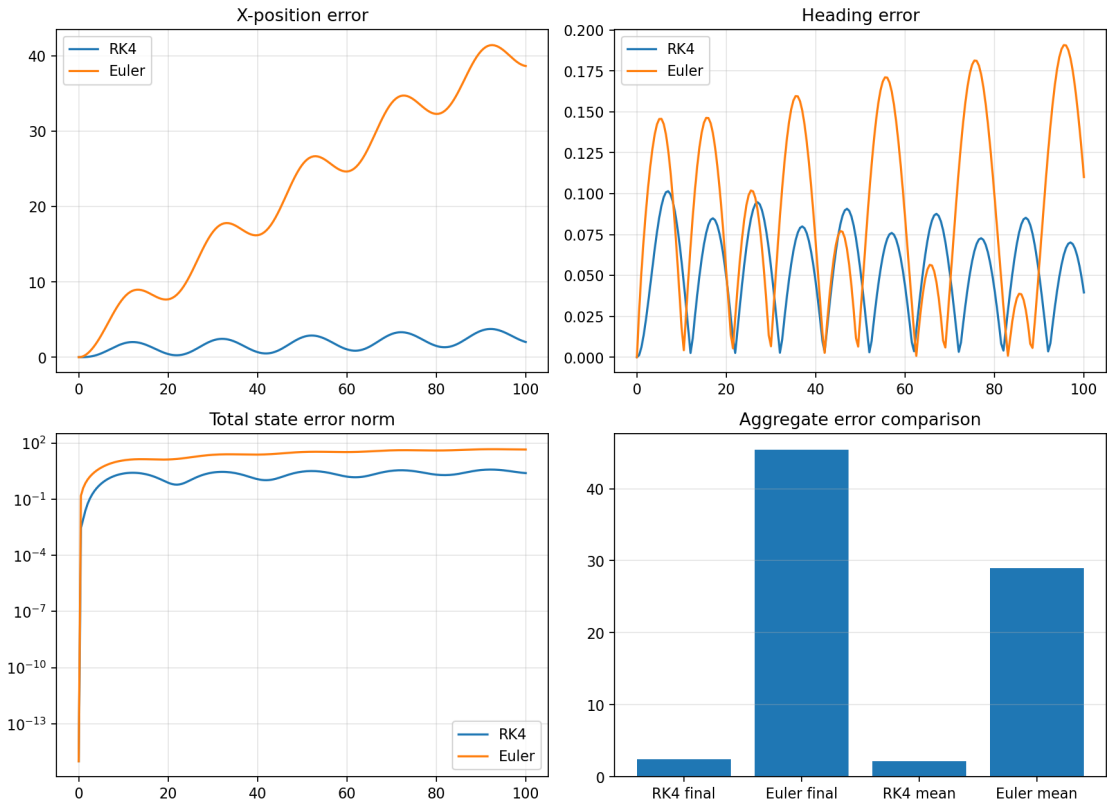

Fig. 19. Recreated RK4-versus-Euler benchmark using the project dynamics implementation.

The framework nevertheless remains simplified. The aircraft model is planar, the LQR design uses a first-order discretization, and the estimator is not yet closed around the controller. These assumptions are acceptable for coursework demonstration but limit physical fidelity and robustness claims.

Future work could close the loop around estimated states, incorporate disturbances such as wind, and extend the design toward uncertainty-aware formulations including stochastic, robust, and distributionally robust optimization[12–17].

6 Conclusion

This report presented a planar trajectory-tracking simulation for the Boeing 747, combining LQR feedback with RK4 integration and a diagnostic Bayes filter. The baseline experiment achieved a steady-state cross-track error below 1 m on a 120 km orbit, with airspeed held within 0.3 m/s of the reference. The framework exercises the core techniques covered in AAE6202—optimisation, numerical ODEs, and probabilistic estimation—within a single integrated application.

The model remains limited to horizontal flight and assumes perfect state feedback. Incorporating altitude dynamics and closing the loop around the filtered state would be natural extensions; the modular code structure should make either step straightforward.

7 One-Page Summary of Group Contributions

Table 6
Group contribution summary

Group Member	Student ID	Specific Components	Deliverables / Evidence	Overall Contribution (%)
Yufa You	25144639R	System architecture, software implementation, report drafting	Simulation framework, code modules, initial manuscript	50
Ruige Yang	25145125R	Mathematical modeling, algorithm derivation, experimental analysis	Theoretical formulations, numerical experiments, result validation	50

Yufa You was responsible for system architecture, software implementation, and report drafting. Ruige Yang was responsible for mathematical modeling, algorithm derivation, and experimental analysis. Topic selection, controller design, and manuscript revision were conducted jointly.

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